

An Exploratory Analysis of Recorded Crime in South Wales: October 2025

Data origin and acquisition

The data used in this analysis was obtained from the UK Police open data portal (data.police.uk), which provides publicly available records of police-recorded crime. The dataset was accessed following the guidance provided in the module document *EDAWork2_AccessingPoliceRecordedIncidentsData_2025_2026*. Using the “Data” tab on the portal, South Wales Police was selected, and the available street-level crime data were downloaded as a compressed file. The street crime file for October 2025 was used for this assignment.

This data set records individual crime incidents and includes information on crime type, outcome, approximate location, geographic coordinates, and small-area geography.

Data formatting and preparation

The original dataset (2025-10-south-wales-street.csv) contained several variables that showed no variation across records. All observations related to incidents recorded in October 2025, and both the “Reported by” and “Falls within” fields consistently referred to South Wales Police. As these variables did not contribute additional analytical insight, they were removed from the dataset. The Context field was also excluded, as it was entirely blank across all observations.

The original LSOA name variable was transformed into a Local Authority variable to support analysis at a broader geographic level. This was achieved by extracting the local authority name from the original LSOA label and discarding the more granular code-based component. Longitude and latitude were initially present in the dataset but were later removed, as spatial mapping was not pursued in the final analysis.

All local authorities present in the dataset were retained for the univariate analysis in order to provide a complete overview of the distribution of recorded incidents across areas. However, for the bivariate analysis examining crime types across local authorities, several authorities with very small numbers of observations (including Torfaen, Pembrokeshire, North Somerset, Manchester, Carmarthenshire, Powys, and Newport) were excluded. This was done to improve the interpretability of the visualisations and to focus the analysis on authorities with sufficient data for meaningful comparison.

Peculiarities and data issues

Several data issues were identified during inspection of the dataset. A number of records were labelled as “No location”, which originally resulted in missing longitude and latitude values. In addition, a substantial number of observations did not contain a Crime ID. These records were retained, as other variables were populated and they still contributed to the exploratory analysis. These features reflect known limitations of police-recorded data and are considered when interpreting the results.

Methodology

Overview

This analysis uses exploratory data analysis techniques to examine police-recorded crime data from South Wales Police for October 2025. The focus is on identifying patterns in crime volume, distribution, and outcomes across offence types and local authorities. The analysis is primarily univariate and bivariate, supported by visualisation to aid interpretation. Limited multivariate exploration was considered, but the short time window and high imbalance across categories restricted its interpretive value.

Univariate Analysis

Univariate analysis was used to examine individual variables in isolation in order to understand their overall distributions and identify dominant categories, rare events, and skewness in the data.

Counts and frequency distributions were produced for key variables, including:

- Crime type
- Local authority
- Last outcome category

Bar charts were used to visualise these distributions, allowing direct comparison of incident volumes across categories. This approach made it possible to identify which crime types and outcomes were most prevalent during October 2025, as well as those that occurred infrequently.

All local authorities present in the dataset were retained for the univariate analysis. This allowed a complete view of how recorded crime was distributed geographically before any filtering was applied for more focused comparisons.

Bivariate Analysis

Bivariate analysis was conducted to explore relationships between pairs of categorical variables and to assess how crime patterns varied across different contexts.

Two main bivariate relationships were examined:

- Crime type by local authority
- Crime type by outcome category

Stacked bar charts were used to visualise these relationships. This made it possible to compare not only total crime volumes but also the composition of crime types or outcomes within each category.

For the analysis of crime types across local authorities, a subset of **major local authorities** was used. Smaller authorities and those with very low incident counts were excluded to avoid distortion. This filtering was applied only to the bivariate analysis; the univariate analysis retained all authorities to preserve a complete overview of the dataset.

Category ordering was based on total incident counts to improve readability and to highlight dominant patterns.

Hypotheses guiding the analysis

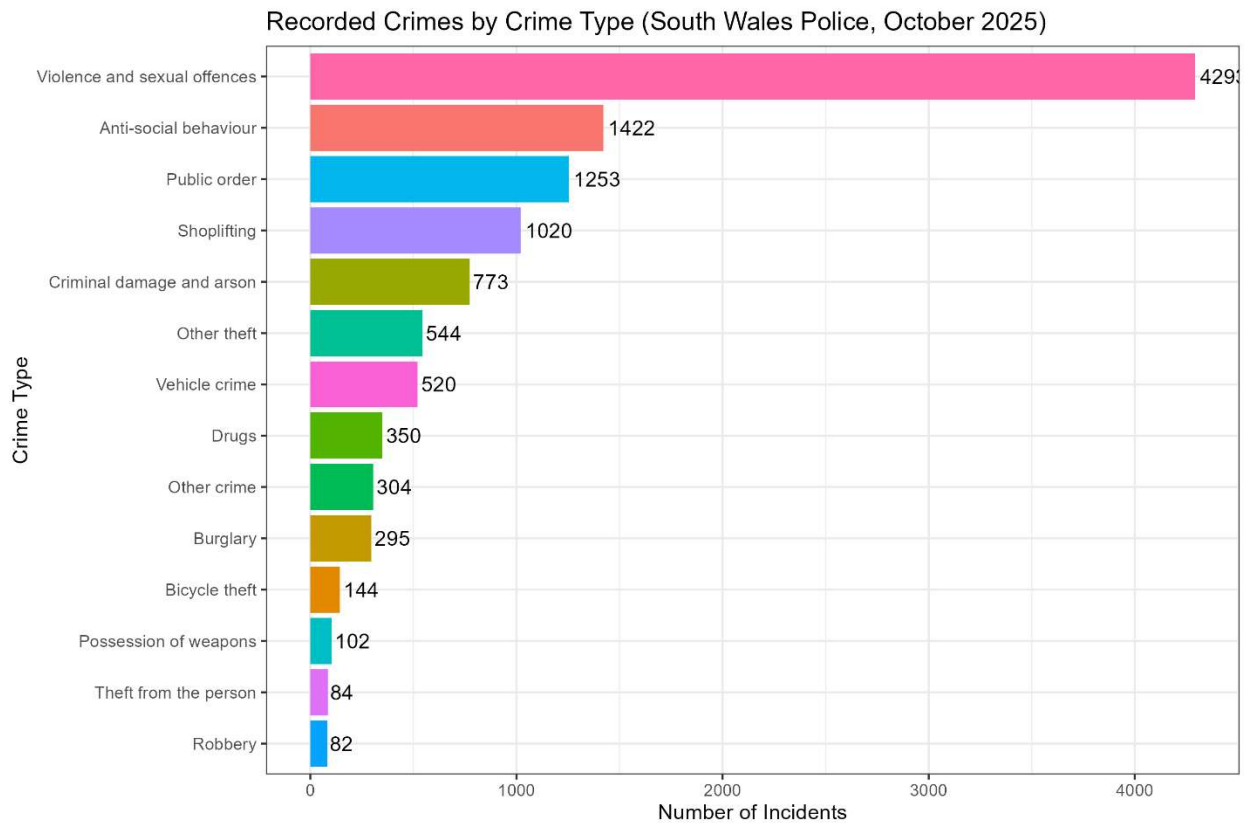
Before conducting the analysis, several expectations were formed based on prior literature and common patterns in police-recorded crime data:

- Certain crime types, particularly violence-related offences and anti-social behaviour, were expected to dominate recorded incidents.
- Crime volumes were expected to be unevenly distributed across local authorities, with urban areas recording higher counts.
- Crime outcomes were expected to vary by offence type, with violent crimes more likely to remain under investigation or unresolved.
- Relationships between crime type, outcome, and location were expected to be visible at an aggregated level, even within a single month of data.

The analysis aimed to assess whether these expectations were supported by the October 2025 South Wales Police data.

Univariate analysis: trends and outliers

Crime type distribution



```
u1_data <- data_major %>%  
  count(`Crime type`) %>%  
  arrange(desc(n))
```

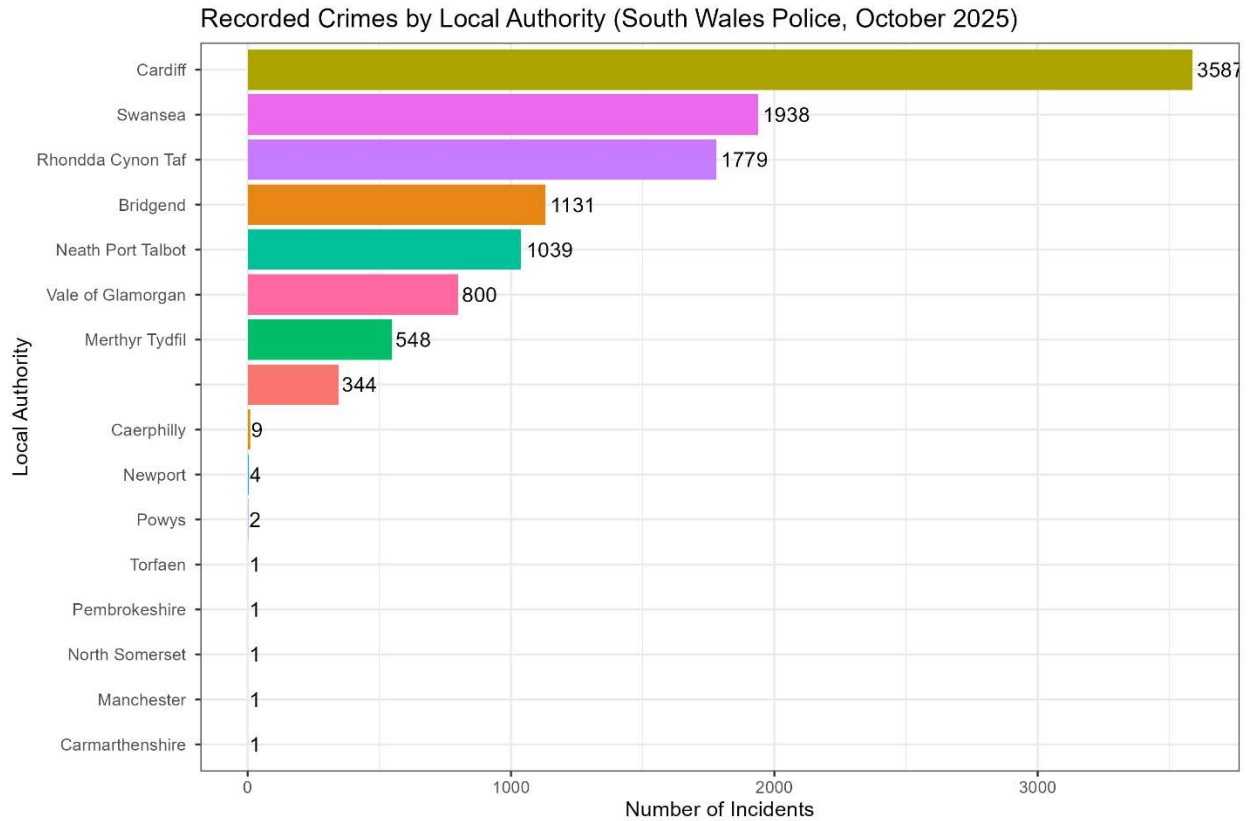
```
u1 <- ggplot(
  u1_data,
  aes(x = reorder(`Crime type`, n), y = n, fill = `Crime type`)
) +
  geom_col(show.legend = FALSE) +
  geom_text(aes(label = n), hjust = -0.1) +
  coord_flip() +
  labs(
    title = "Recorded Crimes by Crime Type (South Wales Police, October 2025)",
    x = "Crime Type",
    y = "Number of Incidents"
  ) +
  theme_bw()
ggsave("univariate_crime_type.png", u1, width = 9, height = 6)
```

The univariate analysis of crime type shows a highly skewed distribution. Violence and sexual offences dominate the dataset by a substantial margin, far exceeding all other categories. Anti-social behaviour, public order offences and shoplifting form a sort of second tier, while crimes such as robbery and theft from the person occur relatively infrequently.

This pattern aligns with expectations for police-recorded data, where high-harm offences and public-order related incidents tend to generate higher reporting volumes. The sharp drop-off between the most common and least common categories highlights strong imbalance in offence frequencies.

A clear outlier is violence and sexual offences, which appear at a much higher frequency than any other category.

Local authority distribution



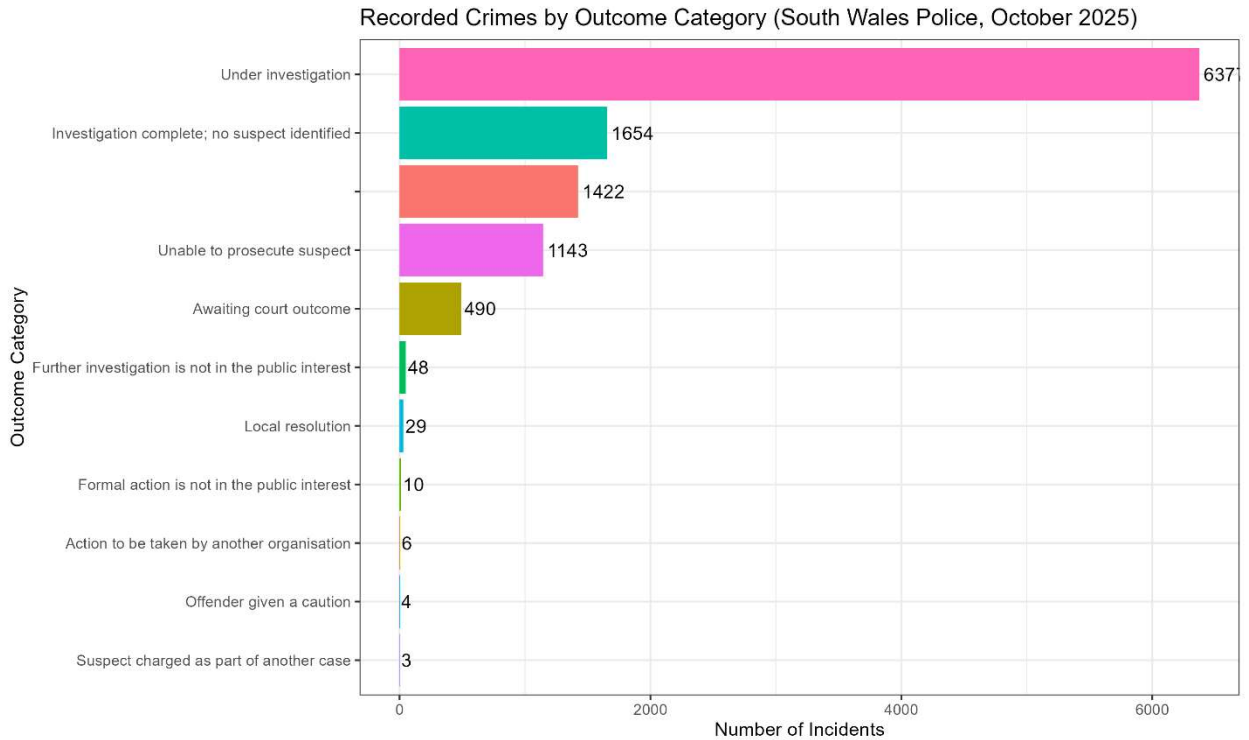
```
u2_data <- data_major %>%
  count(`Local Authority`) %>%
  arrange(desc(n))
```

```
u2 <- ggplot(
  u2_data,
  aes(x = reorder(`Local Authority`, n), y = n, fill = `Local Authority`)
) +
  geom_col(show.legend = FALSE) +
  geom_text(aes(label = n), hjust = -0.1) +
  coord_flip() +
  labs(
    title = "Recorded Crimes by Local Authority (South Wales Police, October 2025)",
    x = "Local Authority",
    y = "Number of Incidents"
  ) +
  theme_bw()
ggsave("univariate_local_authority.png", u2, width = 9, height = 6)
```

Crime counts by local authority show strong geographic concentration. Cardiff records the highest number of incidents, followed by Swansea and Rhondda Cynon Taf. These areas account for the majority of recorded crimes in the dataset.

Several local authorities record extremely low counts, including Torfaen, Pembrokeshire, North Somerset, Manchester, Carmarthenshire, Powys, and Newport. These appear as clear low-end outliers. Their inclusion is informative in the univariate context, as it illustrates the uneven spatial coverage of the dataset, but their small counts limit interpretability in comparative analysis.

Outcome category distribution



```
u3_data <- data_major %>%
  count(`Last outcome category`) %>%
  arrange(desc(n))
u3 <- ggplot(
  u3_data,
  aes(
    x = reorder(`Last outcome category`, n),
    y = n,
    fill = `Last outcome category`
  )
) +
  geom_col(show.legend = FALSE) +
```

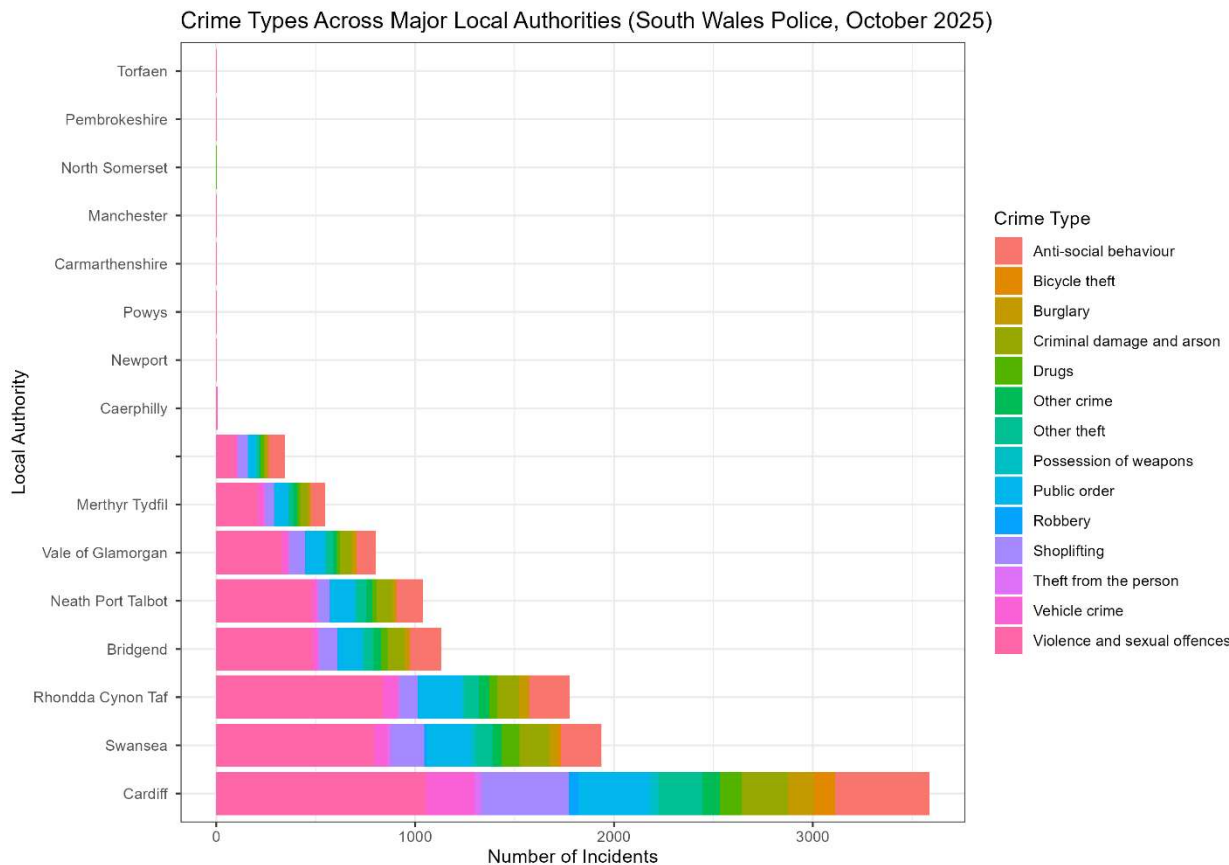
```
geom_text(aes(label = n), hjust = -0.1) +
coord_flip() +
labs(
title = "Recorded Crimes by Outcome Category (South Wales Police, October 2025)",
x = "Outcome Category",
y = "Number of Incidents"
) +
theme_bw()
ggsave("univariate_outcome.png", u3, width = 10, height = 6)
```

Outcome data are highly skewed towards unresolved cases. The dominant category is **“Under investigation,”** followed by **“Investigation complete; no suspect identified.”** A substantial number of records contain no outcome information, forming the third largest group and indicating incomplete case progression at the time of reporting. Formal justice outcomes appear rarely, with **“Unable to prosecute suspect”** being the most common among them.

This suggests that, within the October 2025 snapshot, a large proportion of cases remain unresolved or closed without judicial action. This pattern is consistent with known limitations of police-recorded outcome data, particularly within short time windows.

Bivariate analysis: associations between variables

Crime type by local authority



```
authority_order <- data_major %>%
count(` Local Authority` ) %>%
arrange(desc(n)) %>%
pull(` Local Authority` )
b1_data <- data_major %>%
mutate(` Local Authority` = factor(` Local Authority`, levels = authority_order))

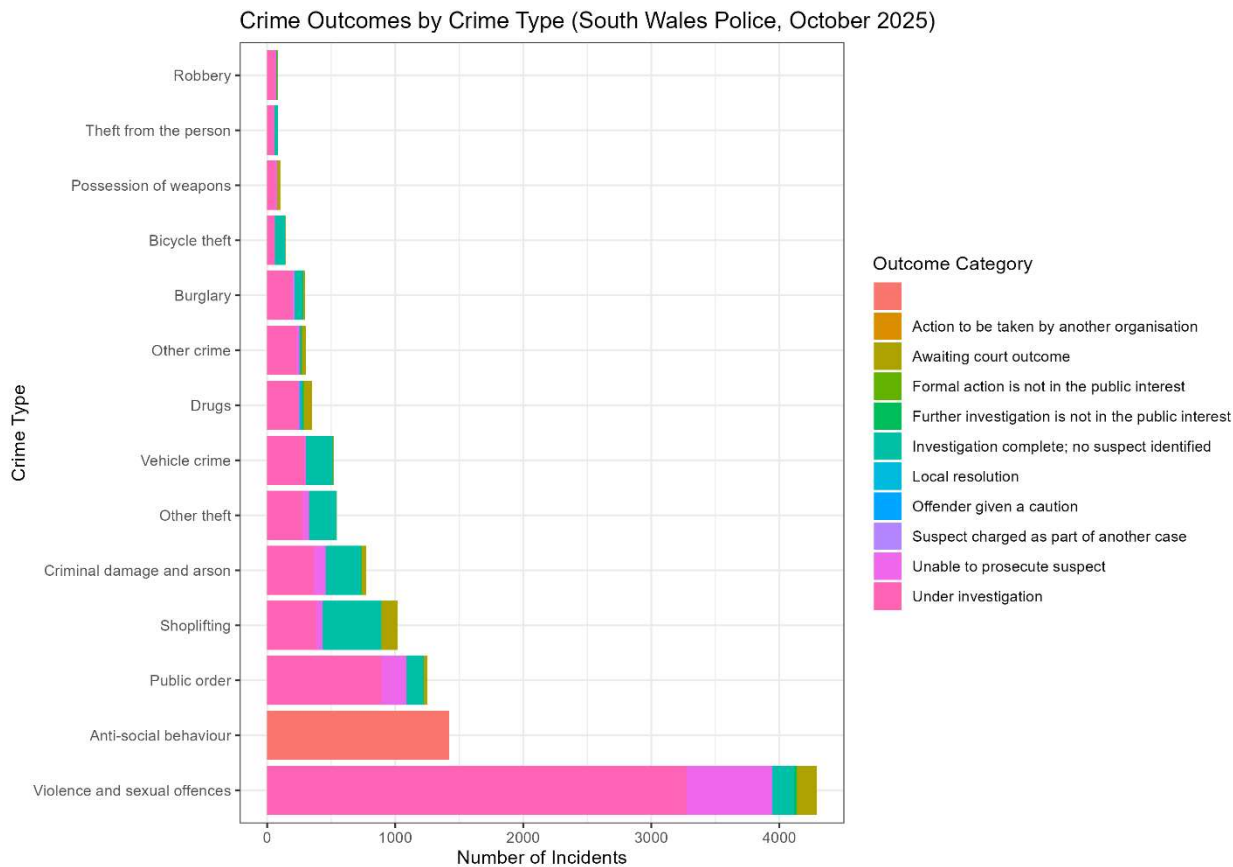
b1 <- ggplot(
b1_data,
aes(x = ` Local Authority`, fill = ` Crime type` )
) +
geom_bar() +
coord_flip() +
labs(
title = "Crime Types Across Major Local Authorities (South Wales Police, October 2025)",
x = "Local Authority",
y = "Number of Incidents",
fill = "Crime Type"
) +
theme_bw()
ggsave("bivariate_authority_crime.png", b1, width = 10, height = 7)
```

The bivariate analysis comparing crime type across major local authorities reveals that differences between areas are driven more by overall volume than by fundamentally different crime profiles.

Across Cardiff, Swansea, Rhondda Cynon Taf, Bridgend, and Neath Port Talbot, the relative composition of crime types remains similar. Violence and sexual offences dominate in every major authority, followed by anti-social behaviour and public order offences.

This indicates a shared pattern of recorded crime across urban areas, rather than distinct local specialization.

Crime outcomes by crime type



```
crime_order <- data_major %>%
  count(`Crime type`) %>%
  arrange(desc(n)) %>%
  pull(`Crime type`)
b2_data <- data_major %>%
  mutate(`Crime type` = factor(`Crime type`, levels = crime_order))

b2 <- ggplot(
  b2_data,
  aes(x = `Crime type`, fill = `Last outcome category`)
) +
  geom_bar() +
  coord_flip() +
  labs(
    title = "Crime Outcomes by Crime Type (South Wales Police, October 2025)",
    x = "Crime Type",
    y = "Number of Incidents",
    fill = "Outcome Category"
  ) +
  theme_bw()
ggsave("bivariate_crime_outcome.png", b2, width = 10, height = 7)
```

This plot highlights meaningful associations between offence type and outcome. Violent offences show a higher absolute number of cases remaining under investigation or unresolved, reflecting their complexity and longer investigative timelines.

Lower-volume crimes, such as robbery or possession of weapons, show fewer total outcomes overall, making patterns harder to interpret reliably. Across most offence types, resolved outcomes remain rare relative to unresolved ones, reinforcing the univariate finding of outcome imbalance.

Negative and limited findings

Several analytical approaches did not reveal strong or unexpected patterns.

Despite examining crime type by local authority, there was no clear evidence of localised crime specialisation among the major authorities. Crime composition appeared broadly similar across areas, suggesting limited geographic differentiation within the single month analysed.

Further analysis of outcomes highlighted important limitations. All records classified as **anti-social behaviour** lacked an associated outcome category, meaning this offence type could not be meaningfully included in outcome-based comparisons. In addition, a substantial number of records across other crime types contained blank or missing outcome values. These gaps reduced the ability to draw strong conclusions about resolution patterns and constrained deeper multivariate exploration.

Attempts at more complex multivariate interpretation were also limited by the **short temporal scope** of the data. As the dataset covers **October 2025 only**, there is no variation across time, preventing the identification of trends, seasonality, or change in outcomes. This single-month snapshot restricts inference to descriptive relationships rather than broader structural dynamics.

Finally, several local authorities and crime categories occurred at very low frequencies. These sparse groups did not provide stable or reliable patterns for inference and were therefore excluded from some bivariate analyses. Their presence reinforces the uneven distribution of recorded crime within the dataset.

Ethical considerations

Several ethical issues are relevant to this analysis of police-recorded crime data.

First, the dataset represents **administrative records rather than confirmed criminal behaviour**. Crimes are recorded based on reports to the police, which means the data reflect reporting practices, public trust in policing, and police recording standards rather than the true prevalence of crime. Interpreting counts as direct measures of criminality could therefore misrepresent communities or offence types.

Second, the analysis aggregates crime at the **local authority level**, which helps reduce disclosure risk but still raises concerns about **stigmatisation**. Areas with higher recorded crime counts may be perceived negatively, even though differences may be driven by population size, reporting behaviour, or policing priorities. Care has been taken to avoid causal claims or comparisons that imply blame or failure of specific communities.

Third, outcome data introduce ethical challenges. A large proportion of cases are listed as *under investigation* or *no suspect identified*, and some records contain no outcome at all. Presenting these outcomes without context could imply inefficiency or lack of accountability. The analysis therefore treats outcomes descriptively and avoids normative judgements about police performance.

Fourth, although the dataset contains no direct personal identifiers, it originates from incidents involving individuals. Ethical handling requires respecting **privacy and dignity**, even in anonymised form. No attempt was made to re-identify individuals, link records across datasets, or analyse rare combinations of attributes that could risk indirect identification.

Finally, the temporal limitation of the data is ethically relevant. The analysis focuses on **October 2025 only**, which restricts generalisability. Presenting findings as representative of longer-term trends would be misleading. All conclusions are therefore framed as **time-specific observations**, not enduring patterns.

Overall, ethical risks were mitigated by aggregating data, avoiding causal or evaluative claims, acknowledging data limitations, and presenting findings in a balanced and contextualised manner.